

20250630EB_MN (ATP, NCT, TDP43, RNA)

Project: Yi-Shuan Tseng Holt Lab
Author: Yi Shuan Tseng
Status: In progress
Entry Created On: 21 Jul 2025 21:48:15 UTC
Entry Last Modified: 09 Sep 2025 04:18:54 UTC
Export Generated On: 01 Sep 2026 18:32:05 UTC

- 2025年6月30日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年7月2日 星期三
- T2 -> T3
- 2025年7月4日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2025年7月7日 星期一
- T4 -> T5
- 2025年7月9日 星期三
- T5 -> T6
- 2025年7月11日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2025年7月14日 星期一
- T6 -> T7
- Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs

2. Remove the PLO and wash the plate with PBS once

3. Coat the plate with laminine for 1hr.

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add 2mL accutase in 15mL falcon

6. Put the tube into 37C water bath. Flick the cells every 5 min until the accutase went cloudy (~45min this time). (There are many cells attached on the well of 15mL falcon. Will use 50mL next time)

7. pipette the accutase gently and slowly around 20-30 times to break the EBs by checking if there are still visible EBs. (I can still see a lot of undissociated EBs this time, so I take the accutase without visible EBs for the further process.

8. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.

9. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
10. Filter the cell with 40um filter into a new 50mL tube
11. Spin the cell down at 100rcf, 3.5min. Remove the supernatent.
12. Add 1ml of T7 to resuspend the cells thouroughly to avoid clusters.
13. Add more T7 (~5mL) to resuspend the cells and count it.
14. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

coating date ^

	A	B	C	D
1		A	B	C
2	0.01% PLO with Na borate	2025/6/20	2025/6/20	2025/6/27
3	laminin	2025/6/23	2025/6/23	2025/6/28

cell info ^

	A	B	C
1	cell	passage	well#
2	WT11	P3	7

Table1 ^

	A	B
1	run1	volume (mL)
2	accutase	4
3	HI-FBS/CTWM	8

Table3 ✓

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	7.21	97	5	36.05	A	24	250	12	416
3					B	24	250	4	139
4					C	6	2500	6	2080

Plate A

^

	1	2	3	4	5	6
A				92, 5uL in total	110, 12 in total	111, 12 in total
B						
C						
D						

Plate B

^

	1	2	3	4	5	6
A						no virus
B						
C						
D						

Plate C

^

	1	2	3
A			
B			

2025年7月15日 星期二

Change T7 -> T7

2025年7月17日 星期四

T7 -> T8

2025年7月18日 星期五

1/2 T8 -> 1/2 T8

2025年7月21日 星期一

1/2 T8 -> 1/2 T8

Prepare 2.5mM and 0.15mM glucose media with full supplement and 1mM pyruvate only

2025年7月22日 星期二

Treat condition 3 & 4 (24h time point for 1 and 2)

plate A: ~17:35

plate B: ~17:50

For plate B: add 1:2500 SPY595 with the 24h treatment and in remaining wells with BP. (The signal of the DNA was bad with only one hour treatment, and I would like to maximize the signal.

2025年7月23日 星期三

Imaging

Plate A1						
	1	2	3	4	5	6
A				1	1	1
B				2	2	2
C				3	3	3
D				4	4	4

^

Plate B1						
	1	2	3	4	5	6
A						1
B						2
C						3
D						4

^

Plate C1			
	1	2	3
A			
B			

^

Treatment time (all)1																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1			110 (plate A)		111 (plate A)		92 (plate A)		TDP43 (20250623MN)		TDP43 (plateB)			RNA (Plate C)		
2	#		treat time	image time	treat time	image time	treat time	image time	treat JFX	image time	treat SPY	treat JFX	image time	treat time	collect	
3	1	2.5mM glu	10:45	11:45	9:42	10:45										
4	2	0.15mM glu	11:30	12:30	10:16	11:17										
5	3	2.5mM glu 24h	7/22 17:35 SirDNA 17:00	18:00	7/22 17:35 SirDN A 16:00	17:00										
6	4	0.15mM glu 24h	7/22 17:35 SirDNA 17:33	18:41	7/22 17:35 SirDN A 16:35	17:36										
7	1	2.5mM glu					12:17	13:19								
8	2	0.15mM glu					12:30	13:34								
9	3	2.5mM glu					7/22 17:35 new media 15:30	16:30								
10	4	0.15mM glu					7/22 17:35 new media 15:45	16:45								
11	1	2.5mM glu							13:45	14:54						
12	2	0.15mM glu							14:00	15:08						
13	1	2.5mM glu py only							14:15	15:15						
14	2	0.15mM glu py only							14:30	15:40						
15	1 24h	2.5mM glu							14:50	15:57						
16	2 24h	0.15mM glu							15:05	16:20						
17	1	2.5mM glu									7/22 17:50	13:15	14:19			
18	2	0.15mM glu									7/22 17:50	13:35	14:37			
19	3	2.5mM glu									7/22 17:50	18:20	19:20			
20	4	0.15mM glu									7/22 17:50	18:30	19:30			
21	1	2.5mM glu												16:00 7/24	17:00	
22	2	0.15mM glu												16:00 7/24	17:00	
23	3	2.5mM glu												16:00 7/23	17:00	
24	4	0.15mM glu												16:00 7/23	17:00	
25	5	2.5mm + DMSO 6h												11:00	17:00	
26	6	2.5mM glu +2uM MG132 6h												11:00	17:00	

Image (NCT, column 5 and 6)

NCT_O.C. , 60X, 600x900, 16-bit

^

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	20	300 ms	accidentally set a different power... (I used to do 30%)
3	640	16	10	100 ms	
4	488	16	50	1s	

*110 24h 0.15mM: there is a sudden intensity drop from frame 28-33. (truncate them to fit the model)

Image (ATP/ADP ratio with vLH92, column4)

O.C. , 60X, 600x900, 16-bit

^

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	20	200
3	Dapi-ex-Green-em	16	20	300

Image (TDP43, plateB)

Treat cells with 2.5uM JFX650 before the imaging. (0.16uL per well)

dilute 1mM JFX650

--> 1:400 --> 2.5uM

--> 1:2500 --> 1nM (working concentration)

15 positions/well, didn't take movie this time

O.C. , 60X, 600x900, 16-bit1

^

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	20	500ms	
3	640	16	10	300ms	

2025年7月24日 星期四

Collect plate C RNA
11:00 treat A3 and B3
16:00 treat A1,2 and B1,2
17:00 collect

Plate C conditions				^
	1	2	3	
A	2.5mm + DMSO 6h	2.5mM glu 1h	2.5mM glu 24h	
B	2.5mM + 2uM MG132 6h	0.15mM glu 1h	0.15mM glu 24h	

MG132
stock: 10.5mM
working: 2uM
dilution: 5250
0.38 uL in 2mL media